

■ 2.2.16 Defining Numerical Values

When you define a mathematical function, you will often want to give approximate numerical rules for it, as well as exact symbolic ones.

$$N[expr] = value \quad \text{define a numerical value}$$

Defining a numerical value.

This specifies an exact value for $f[1]$.

```
In[1]:= f[1] = ax
Out[1]= ax
```

This defines an approximate numerical value for $f[x_+]$.

```
In[2]:= N[f[x_]] := N[Sum[x^-i/i^2, {i, 20}]]
```

Only the exact definition is automatically used.

```
In[3]:= f[1] + f[6]
Out[3]= ax + f[6]
```

This uses the numerical definition.

```
In[4]:= N[%]
Out[4]= 0.17418 + ax
```

The numerical definition is automatically associated with f , rather than N .

```
In[5]:= ?f
f
f/: f[1] = ax

f/: N[f[x_]] := N[Sum[ $\frac{x^{-i}}{i^2}$ , {i, 20}]]
```